

Word Problems on Polynomial Addition & Subtraction

Algebra 1 (US Common Core) | Homework Package

1 Definitions & Core Concepts

Key Terms

- **Polynomial:** an algebraic expression consisting of variables, coefficients, and non-negative integer exponents combined using addition, subtraction, and multiplication. Example: $3x^2 - 5x + 7$.
- **Standard Form:** a polynomial written in descending order of degree (from highest exponent to lowest).
- **Like Terms:** terms with the exact same variables raised to the exact same powers. Only like terms can be added or subtracted.

Polynomial Addition in Word Problems

Used when combining quantities, finding total perimeters, total revenue, or total combined amounts. Combine like terms by adding their coefficients:

$$(ax^n) + (bx^n) = (a + b)x^n$$

Polynomial Subtraction in Word Problems

Used when finding differences, remaining amounts, profit ($P = R - C$), or missing lengths. Always distribute the negative sign across every term of the polynomial being subtracted:

$$-(ax^2 + bx + c) = -ax^2 - bx - c$$

2 Worked-Out Examples

Example 1: Geometric Application (Addition)

A triangular garden plot has side lengths $P_1 = 3x^2 + 2x - 5$, $P_2 = x^2 - 4x + 8$, and $P_3 = 2x^2 + 5x + 1$ feet. Write a polynomial in standard form for the total perimeter.

Solution. Perimeter is the sum of all sides:

$$\text{Perimeter} = (3x^2 + 2x - 5) + (x^2 - 4x + 8) + (2x^2 + 5x + 1)$$

Group like terms:

$$= (3x^2 + x^2 + 2x^2) + (2x - 4x + 5x) + (-5 + 8 + 1)$$

Combine coefficients:

$$\text{Perimeter} = 6x^2 + 3x + 4 \text{ feet}$$

Example 2: Financial Application (Subtraction)

A company's revenue is $R(x) = 7x^2 + 110x + 500$ dollars and its cost is $C(x) = 3x^2 + 45x + 200$ dollars. Write a polynomial model for profit $P(x)$.

Solution. Profit = Revenue – Cost:

$$P(x) = (7x^2 + 110x + 500) - (3x^2 + 45x + 200)$$

Distribute the negative sign:

$$P(x) = 7x^2 + 110x + 500 - 3x^2 - 45x - 200$$

Group and combine:

$$P(x) = (7x^2 - 3x^2) + (110x - 45x) + (500 - 200) = 4x^2 + 65x + 300 \text{ dollars}$$

3 Homework Practice: 40 Word Problems

Part 1: Geometry & Perimeter Problems (1–10)

1. The three sides of a triangle are $2x^2 + 3x - 1$, $x^2 - 4x + 5$, and $3x^2 + x + 2$. Find the simplified polynomial for the perimeter.
2. A rectangular field has length $4x + 7$ meters and width $2x - 3$ meters. Find the polynomial for the perimeter.
3. Rectangle A has perimeter $6x^2 + 8x + 10$ inches and Rectangle B has perimeter $4x^2 - 3x + 6$ inches. How much greater is A's perimeter than B's?
4. The perimeter of a quadrilateral is $10x^2 + 5x + 12$. Three sides are $2x^2 + 3$, $3x^2 - x + 4$, and $4x^2 + 2x - 1$. Find the fourth side.
5. A square poster has side $3x + 5$ cm; a regular pentagon banner has side $2x + 4$ cm. Write an expression for the sum of their perimeters.
6. Find the difference between the perimeter of a triangle with sides $3x + 2$, $4x - 1$, $5x + 3$, and the perimeter of a square with side $3x + 1$.
7. An outer garden rectangle has length $5x + 2$, width $3x + 1$. An inner pool has length $2x + 1$, width $x + 3$. Find the total combined perimeter.
8. A non-isosceles trapezoid has sides $x^2 + 2x + 1$, $2x^2 - x + 3$, $x^2 + 4$, and $3x^2 - 2x + 5$. Express its total perimeter in standard form.
9. An isosceles triangle has two equal sides $4x^2 - 3x + 2$ and base $2x^2 + 5x - 1$. Write a polynomial for the perimeter.
10. A hexagon has sides $x^2 + 1$, $2x^2 - 3$, $x^2 + 4$, $3x^2 - x$, $2x^2 + 2x$, and $x^2 + 5$. Find the simplified perimeter.

Part 2: Business, Revenue, Cost & Profit Problems (11–20)

11. A software firm's monthly revenue is $R(x) = 5x^2 + 120x + 500$ and operating costs $C(x) = 2x^2 + 45x + 300$. Find profit $P(x) = R(x) - C(x)$.
12. Company A's revenue is $A(x) = 3x^3 + 4x^2 - 10x + 50$; Company B's is $B(x) = x^3 - 2x^2 + 15x + 20$. Find the total combined revenue $T(x)$.
13. Morning sales $M(t) = 4t^2 + 15t + 80$; afternoon sales $A(t) = 3t^2 - 8t + 120$. Find daily sales $S(t) = M(t) + A(t)$.
14. Factory 1 cost $C_1(x) = 0.5x^2 + 20x + 1000$; Factory 2 cost $C_2(x) = 0.8x^2 + 15x + 800$. Express $C_1(x) - C_2(x)$ in standard form.
15. Budget $B(x) = 15x^2 + 200x + 5000$ funds Project 1 ($E_1(x) = 6x^2 + 80x + 1800$) and Project 2 ($E_2(x) = 4x^2 + 50x + 1200$). Write the remaining unspent budget.

16. A plant has fixed overhead \$1500 and variable cost $3x^2 + 25x$. Revenue from selling x units is $7x^2 + 100x$. Write profit $P(x)$ in standard form.
17. Three branches report sales $S_1(x) = 2x^2 + 5x + 10$, $S_2(x) = 3x^2 - 2x + 15$, $S_3(x) = x^2 + 4x - 5$. Find total sales.
18. Adult ticket revenue $A(x) = 8x^2 + 40x$; child ticket revenue $C(x) = 3x^2 + 15x$. Express total ticket revenue.
19. Product Line A profit $P_1(x) = 10x^2 - 30x + 200$; Line B profit $P_2(x) = 6x^2 + 20x - 50$. How much more profit does A generate?
20. Department A expenses $E_A(x) = 4x^2 - 7x + 90$; Department B expenses $E_B(x) = 5x^2 + 12x - 30$. Write the combined total expenses.

Part 3: Science, Physics & Population Problems (21–30)

21. City 1 population $P_1(t) = 5t^2 + 12t + 500$; City 2 population $P_2(t) = 3t^2 - 4t + 350$ (t in years). Write the total combined population.
22. Using Question 21's models, write a polynomial for how many more residents City 1 has than City 2.
23. Rocket A reaches height $h_1(t) = -16t^2 + 120t + 50$ feet; Rocket B reaches $h_2(t) = -16t^2 + 90t + 80$ feet. Find $h_1(t) - h_2(t)$.
24. Block 1 height $y_1(x) = 2x^2 + 3x + 4$ cm; Block 2 height $y_2(x) = x^2 - x + 6$ cm, stacked vertically. Find the total combined height.
25. Tank A holds $V_A(t) = 10t^2 - 15t + 200$ gallons; Tank B holds $V_B(t) = 8t^2 + 20t + 150$ gallons. Write the total volume.
26. Liquid is drained from Tank A ($V_A(t) = 10t^2 - 15t + 200$) by $D(t) = 3t^2 - 5t + 40$ gallons. Find the remaining liquid.
27. Culture 1 has $N_1(t) = 100t^2 + 50t + 500$ cells; Culture 2 has $N_2(t) = 80t^2 - 20t + 300$ cells. Write the total bacteria count.
28. Solar array generates $S(t) = 4t^2 + 30t + 10$ kWh; grid power supplies $G(t) = 2t^2 - 10t + 5$ kWh. Write the total energy provided.
29. Car 1 position $x_1(t) = 3t^2 + 20t + 5$ meters; Car 2 position $x_2(t) = 2t^2 + 25t + 12$ meters. Express the separation distance $x_1(t) - x_2(t)$.
30. Morning temperature $T_1(m) = m^2 + 2m + 60$ degrees; afternoon rise $T_2(m) = 2m^2 - m + 5$. Write the total afternoon temperature.

Part 4: Real-World Applications & Multi-Step Problems (31–40)

31. Male enrolment $B(x) = 3x^2 + 14x + 120$; female enrolment $G(x) = 2x^2 + 18x + 110$. Find total enrolment $E(x) = B(x) + G(x)$.
32. Using Question 31's data, write a polynomial for $G(x) - B(x)$.
33. Library has Fiction $F(x) = 5x^2 - 8x + 400$, Non-fiction $N(x) = 3x^2 + 12x + 250$, Reference $R(x) = x^2 - 2x + 100$. Find the total book count.
34. Corn yield $C(a) = 4a^2 + 30a + 100$ bushels; wheat yield $W(a) = 3a^2 - 10a + 80$ bushels. Express total grain yield in standard form.
35. Using Question 34, write a polynomial for $C(a) - W(a)$.
36. Person A saves $S_1(m) = 50m^2 + 200m + 500$ dollars; Person B saves $S_2(m) = 40m^2 + 150m + 300$ dollars over m months. Find their combined savings.

37. Truck driver covers $d_1(t) = 2t^2 + 15t + 30$ miles Day 1 and $d_2(t) = 3t^2 - 5t + 45$ miles Day 2. Write the total distance.
38. Highway target length $D(t) = 8t^2 + 20t + 100$ meters. Workers complete $d(t) = 5t^2 + 12t + 40$ meters in Month 1. Write the remaining distance.
39. App receives $I(d) = 12d^2 + 50d + 300$ iOS downloads and $A(d) = 15d^2 + 40d + 450$ Android downloads. Write the total downloads.
40. Using Question 39, find the polynomial for $A(d) - I(d)$.

4 Answer Key

1. $6x^2 + 6$
2. $12x + 8$
3. $2x^2 + 11x + 4$
4. $x^2 + 4x + 6$
5. $22x + 40$
6. 0
7. $22x + 14$
8. $7x^2 - x + 13$
9. $10x^2 - x + 3$
10. $10x^2 + x + 7$
11. $P(x) = 3x^2 + 75x + 200$
12. $T(x) = 4x^3 + 2x^2 + 5x + 70$
13. $S(t) = 7t^2 + 7t + 200$
14. $-0.3x^2 + 5x + 200$
15. $5x^2 + 70x + 2000$
16. $P(x) = 4x^2 + 75x - 1500$
17. $6x^2 + 7x + 20$
18. $11x^2 + 55x$
19. $4x^2 - 50x + 250$
20. $9x^2 + 5x + 60$
21. $8t^2 + 8t + 850$
22. $2t^2 + 16t + 150$
23. $30t - 30$
24. $3x^2 + 2x + 10$
25. $18t^2 + 5t + 350$
26. $7t^2 - 10t + 160$
27. $180t^2 + 30t + 800$
28. $6t^2 + 20t + 15$
29. $t^2 - 5t - 7$
30. $3m^2 + m + 65$
31. $E(x) = 5x^2 + 32x + 230$
32. $-x^2 + 4x - 10$
33. $9x^2 + 2x + 750$
34. $7a^2 + 20a + 180$
35. $a^2 + 40a + 20$
36. $90m^2 + 350m + 800$
37. $5t^2 + 10t + 75$
38. $3t^2 + 8t + 60$
39. $27d^2 + 90d + 750$
40. $3d^2 - 10d + 150$